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Remarkably high apparent quantum yield of the overall photocatalytic H₂O splitting achieved by utilizing Zn ion added Ga₂O₃ prepared using dilute CaCl₂ solution†

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Remarkably high photocatalytic activity for the overall H₂O splitting, where the activity was 32 mmol h^{−1} for H₂ production and 16 mmol h^{−1} for O₂ production under irradiation from a 450 W high-pressure Hg lamp and the apparent quantum yield (AQY) was 71% under irradiation at 254 nm, was achieved by utilizing a Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃ photocatalyst when Ga₂O₃ was prepared using dilute CaCl₂ aqueous solution having a concentration of 0.001 mol l^{−1}.

Gallium oxide is reported to be one of the simple oxide photocatalysts that exhibit photocatalytic activity for the overall H₂O splitting and has been extensively studied.^{1–9} Although the band gap of this photocatalyst is too wide (4.4 eV evaluated from the UV spectrum) to carry out a photocatalytic reaction under visible light irradiation, the photocatalyst has sufficiently high potential for the overall photocatalytic H₂O splitting though the photocatalytic activity itself is low. Therefore, various investigations, such as examination of the influence of the preparation conditions of Ga₂O₃ and modification of the surface and the bulk, have been performed in order to improve photocatalytic performances.^{2,4,6} As a result, Zn ion added Ga₂O₃ combined with a Rh_yCr_{2–y}O₃ co-catalyst has been reported to exhibit extremely high photocatalytic activity in the overall H₂O splitting, where the preferable state of the photocatalyst was Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃ and the activity was 21 mmol h^{−1} and 10.5 mmol h^{−1} for H₂ and O₂, respectively, under irradiation from a 450 W high-pressure Hg lamp in an inner-irradiation type photoreaction cell.⁴ The activity has been confirmed to be at the same level as that in a NiO/La-NaTaO₃ photocatalyst reaction carried out under nearly the same conditions, and this has been reported to have the highest efficiency in the overall H₂O splitting.^{10,11} However, the detailed quantum yield and the potential to further improve the photocatalytic activity of the overall H₂O splitting have not been investigated so far.

From our previous investigations into improvement of the photocatalytic performance of Ga₂O₃ to H₂O splitting,⁴ the following two particular points are important; one is the control of the surface states by combining an effective co-catalyst for suppressing the reverse chemical process that occurs over the surface and the other is the modification of the bulk state to prevent the recombination and the mobility of electrons and holes that are produced by irradiation of the bulk of the photocatalyst. The former was achieved by the combination of a Rh_yCr_{2–y}O₃ co-catalyst, and the latter was achieved by the addition of a small amount of metal ions such as Ca, Zn, Cr, Sr, and Ba ions, in particular, Zn ions to Ga₂O₃. With respect to the latter, Ga₂O₃ containing a small amount of metal ions that have a positive influence on the improvement of the photocatalytic activity has the potential to further improve the activity for H₂O splitting.

In this study, in order to prepare an effective Ga₂O₃ photocatalyst for the overall H₂O splitting, we investigated the influence of the preparation conditions on the photocatalytic performance. In particular, we studied the influence of small amounts of metal ions in the preparation atmosphere by the ammonia precipitation method on photocatalytic performance based on the latter point described above. In this respect, we investigated the influence of the co-existence of small amounts of Ca ions, which are also reported to be the second effective additive for improving the photocatalytic activity of Ga₂O₃,² in the preparation atmosphere of the Ga₂O₃ photocatalyst on the overall photocatalytic H₂O splitting. Here, we report on the influence of Ca ions in the preparation of Ga₂O₃ and the apparent quantum yields of the photocatalytic reaction over various photocatalysts.

The Ga₂O₃ photocatalyst was prepared by an ammonia precipitation method by using distilled and de-ionized water (supplied from EYELA Still Ace SA-2100 E1, resistivity 12 MΩ cm at 298 K) and aqueous solutions of CaCl₂ dissolved in ultra-pure water (supplied from Millipore, resistivity 18.2 MΩ cm at 298 K). Ga(NO₃)₃·*n*H₂O (99.9%, obtained from High Purity Chemicals) (12.6 g) was dissolved in the de-ionized water or various concentrations of CaCl₂ ultra-pure water solution (200 ml). Hydroxylation was carried out by dripping ammonium hydroxide solution to

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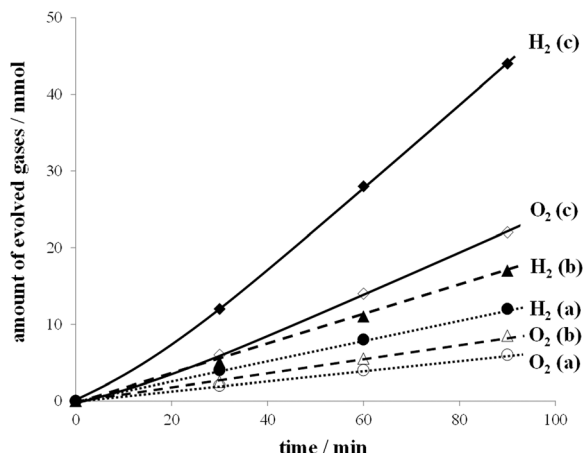


Fig. 1 Photocatalytic evolution of H₂ and O₂ in time over (a) Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Ga₂O₃(DI), (b) Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Ga₂O₃(UP-Ca(0.001)), and (c) Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca(0.001)).

have a final pH value of 9. The produced hydroxide was filtered and calcined at 1273 K for ten hours to obtain Ga₂O₃. Here, Ga₂O₃ prepared by using the de-ionized water is denoted as Ga₂O₃(DI) and that prepared by using CaCl₂ ultra-pure water solution is denoted as Ga₂O₃(UP-Ca(concentration per mol per l)). The details of the addition of Zn ions and the combination of a Rh_{0.5}Cr_{1.5}O₃ co-catalyst are provided in previous reports.^{4,6}

Fig. 1 shows the evolution of H₂ and O₂ in time by the overall photocatalytic H₂O splitting over (a) Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Ga₂O₃(DI), (b) Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Ga₂O₃(UP-Ca(0.001)), and Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca(0.001)). Photocatalytic H₂O splitting was performed in an inner-irradiation type photoreaction cell connected to an isovolumetric closed gas circulation system (dead volume 1.7 l) equipped with a vacuum line and a gas chromatograph sample inlet. The prepared photocatalyst (1 g) was suspended in thoroughly outgassed ultra-pure water (0.65 l) by magnetic stirring in the cell and the suspension was irradiated using a high-pressure Hg lamp (USHIO UM-452, 450 W). The evolved H₂ and O₂ were collected in a sampling tube and analyzed by a gas chromatograph.

As shown in Fig. 1, H₂ and O₂ are produced constantly with relatively high activity from the beginning of irradiation over all photocatalysts. Particularly, photocatalytic activity improved significantly by applying Ga₂O₃(UP-Ca(0.001)) as the photocatalyst. Moreover, remarkably high photocatalytic activity can be confirmed if Ga₂O₃(UP-Ca(0.001)) with added 3 mol% of Zn ions is applied as the photocatalyst. Then, the photocatalytic properties of Zn doped Ga₂O₃ (UP-Ca(0.001)) in the overall H₂O splitting were examined in detail.

Fig. 2 shows the evolution of H₂ and O₂ in time by the overall photocatalytic H₂O splitting over Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca(0.001)). As shown in Fig. 2, H₂ and O₂ were produced in the stoichiometric ratio with remarkably high photocatalytic activity and the pressure of the system became 1 atmosphere within 1.5 hours at the first run. Thereafter, the photocatalytic reaction was carried out for 80 minutes followed by the evacuation of the system and the reaction was

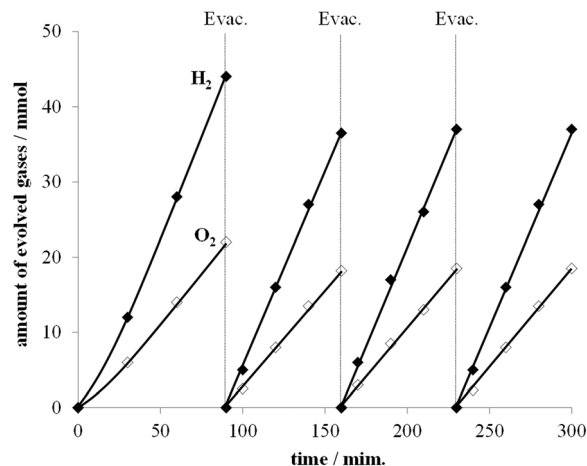


Fig. 2 Photocatalytic evolution of H₂ and O₂ in time over Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca(0.001)).

restarted. From the result in Fig. 2, it is also confirmed that H₂ and O₂ were produced with remarkably high activity if the reaction was restarted repeatedly. The values of the activity in a steady state were 32 mmol h⁻¹ and 16 mmol h⁻¹ for H₂ and O₂, respectively. These values are extremely high compared with the activity of Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(DI), where the reaction was carried out in the same reaction system in the present study, which has been reported previously.⁴ Accordingly, the effect of the concentration of CaCl₂ in the preparation solution of Ga₂O₃ on the photocatalytic activity of Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca) was examined.

Fig. 3 shows the photocatalytic activity of Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca) in the overall photocatalytic H₂O splitting as a function of the concentration of CaCl₂ in the preparation solution. As shown in Fig. 3, the photocatalytic activity depends on the concentration of CaCl₂ in the preparation solution. In particular, the activity improves with the increase of the concentration of CaCl₂ and becomes the maximum at 0.001 mol l⁻¹, while the activity decreases as the concentration

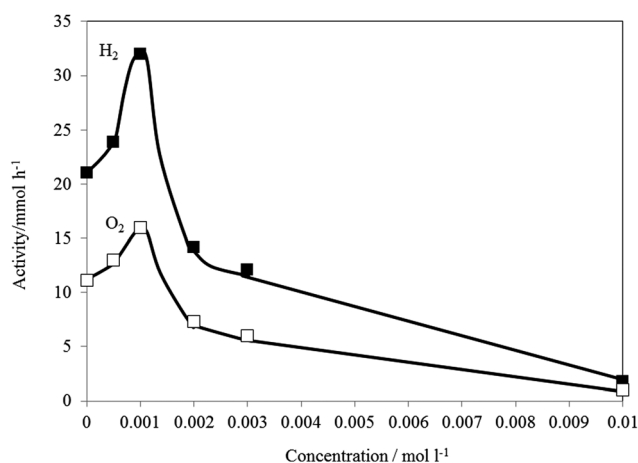


Fig. 3 Photocatalytic activity for the overall H₂O splitting on Rh_{0.5}Cr_{1.5}O₃(Rh; 0.5 wt%)/Zn(3 mol%)-Ga₂O₃(UP-Ca) as a function of the concentration of CaCl₂ in the solution for preparing Ga₂O₃.

is increased above 0.001 mol l^{-1} . From the result in Fig. 3, it is noticed that application of Ga_2O_3 prepared using the dilute CaCl_2 solution by the ammonia precipitation method is effective for the photocatalytic activity of Zn added Ga_2O_3 in the overall H_2O splitting. In this case, the Ca ion in the preparation atmosphere probably positively influences the photocatalytic performance of Zn added Ga_2O_3 .

Accordingly, the state of $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$, which was the preferable Ga_2O_3 of the Zn added Ga_2O_3 photocatalyst for the overall H_2O splitting as shown in Fig. 3, was investigated. The crystallography and the morphology of Ga_2O_3 were examined by XRD (Rigaku RINT-2200) and SEM (JEOL JSM-7600F). No significant change could not be observed in the XRD pattern and the morphology of the Ga_2O_3 particles of $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ in the SEM photograph compared with those of $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0))$ (see Fig. S1 and S2 in the ESI†). The energy dispersive X-ray emission spectrum (EDS) was also measured. From the result of EDS, the spectrum attributed to Ca shows a very small peak (see Fig. S3 in the ESI†). Then, the content of Ca ions in the prepared $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ was examined by inductively coupled plasma optical emission spectroscopy (ICP-OES; Perkin Elmer Optima 4300DV). From the result of ICP, the amount of Ca ions in $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ was 0.89 mol% to Ga ions. From these results, it is noticed that a small amount of Ca ions may disperse homogeneously in the bulk of $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$. Therefore, the influences of the added Ca ion in the bulk were difficult to observe clearly in the XRD pattern and the SEM photograph. On the other hand, the $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ photocatalyst was prepared by the calcination of a hydroxide precursor, where the small amount of Ca ions dispersed homogeneously in gallium hydroxide, at 1273 K. This means that the added Ca ions strongly interacted with Ga_2O_3 bulk by calcination to be in a stable state in the $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ photocatalyst. However, further examinations are necessary to learn the state of the Ca ion in the Ga_2O_3 bulk and the influences on the overall H_2O splitting in detail.

The photocatalytic activities of various Ga_2O_3 and Zn added Ga_2O_3 combined with $\text{Rh}_{0.5}\text{Cr}_{1.5}\text{O}_3(\text{Rh}; 0.5 \text{ wt}\%)$ as the co-catalyst for the overall H_2O splitting are summarized in Table 1. Table 1 shows the photocatalytic activity as the production rate of H_2 and O_2 under irradiation using a 450 W high-pressure Hg lamp in an inner-irradiation type photoreaction cell and the apparent quantum yield (AQY). The apparent quantum yield (AQY) of photocatalytic H_2O splitting was measured by using a top-irradiation type photoreaction cell connected to a closed gas circulation system (dead volume 0.4 l). Irradiation was carried out using a 450 W high-pressure Hg lamp through a water-cooling jacket made of quartz, a 20 mm diameter slit and a bandpass filter ($\lambda = 254 \text{ nm}$, obtained from Edmund Optics). The incident photocurrent was measured by using a calibrated silicon photodiode (made by Hamamatsu Photonics) at the level of the water-gas interface, and photon flow was calculated by the photocurrent. The AQY was expressed as a percentage of the number of photons used for H_2O splitting to the number of incident photons.

As shown in Table 1, the photocatalytic activity of Ga_2O_3 noticeably improves if the photocatalyst is prepared using dilute CaCl_2 solution. This reflects the influence of Ca ion

Table 1 Photocatalytic performance of Ga_2O_3 and Zn (3 mol%) doped Ga_2O_3 prepared with de-ionized water and dilute CaCl_2 ultra-pure water solution (0.001 mol l^{-1}) in the overall H_2O splitting

Photocatalysts ^a	Activity/ mmol h^{-1} ^b		
	H_2	O_2	AQY at 254 nm/%
$\text{Ga}_2\text{O}_3(\text{DI})^6$	7.9	3.9	24
$\text{Ga}_2\text{O}_3(\text{UP-Ca}0.001)$	11	6	—
$\text{Zn}(3 \text{ mol}\%)-\text{Ga}_2\text{O}_3(\text{DI})^4$	21	10.5	57
$\text{Zn}(3 \text{ mol}\%)-\text{Ga}_2\text{O}_3(\text{UP-Ca}0.001)$	32	16	71

^a Photocatalysts are used in combination with the co-catalyst of $\text{Rh}_{0.5}\text{Cr}_{1.5}\text{O}_3(\text{Rh}; 0.5 \text{ wt}\%)$. $\text{Ga}_2\text{O}_3(\text{DI})$ is Ga_2O_3 prepared with de-ionized water and $\text{Ga}_2\text{O}_3(\text{UP-Ca}0.001)$ is Ga_2O_3 prepared with CaCl_2 ultra-pure water solution having a concentration of 0.001 mol l^{-1} .

^b Photocatalytic activity is evaluated in an inner-irradiation type photoreaction cell under irradiation from a 450 W high-pressure Hg lamp.

doping in the bulk of the Ga_2O_3 photocatalyst if Ga_2O_3 is prepared using dilute CaCl_2 solution as we discussed above. Moreover, remarkably high photocatalytic activity is achieved on the Zn added Ga_2O_3 if Ga_2O_3 is prepared by applying a dilute CaCl_2 solution. The activity is extremely higher than that of the Zn added Ga_2O_3 photocatalyst prepared by the method referred to in our previous report. These results also suggest that Ca ions in the preparation atmosphere positively influence the prepared Ga_2O_3 photocatalyst itself in the overall H_2O splitting. Then, the AQY of the overall photocatalytic H_2O splitting under irradiation at 254 nm was evaluated. The results are also listed in Table 1. As shown in Table 1, the values of the AQY are correlated with the corresponding photocatalytic activities. From the results, it is noticed that the values of AQY of the overall H_2O splitting over the Zn added Ga_2O_3 photocatalysts exceed 50%. Particularly, the value of the Zn- $\text{Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ photocatalyst is 71% which is extremely higher than that of NiO/La-NaTaO_3 , which has been reported as being the photocatalyst showing the highest AQY in the overall H_2O splitting.^{8,9}

The results obtained in this study clearly demonstrate that the addition of a very small amount of CaCl_2 under the preparation atmosphere makes the photocatalytic performances of Ga_2O_3 itself in the overall H_2O splitting improve further. Particularly, the photocatalytic activity also improved by the addition of Zn ions and the extremely high activity and AQY can be observed as shown in Table 1. The control of the addition of CaCl_2 in the preparation atmosphere of Ga_2O_3 is one of the significant factors to improve the photocatalytic activity for the overall H_2O splitting.

In conclusion, it was confirmed that the photocatalytic properties of $\text{Rh}_{0.5}\text{Cr}_{1.5}\text{O}_3/\text{Ga}_2\text{O}_3$ in the overall H_2O splitting were improved by applying Ga_2O_3 prepared by using a dilute CaCl_2 solution by the ammonia precipitation method. Moreover, further improvement of photocatalytic performance in H_2O splitting was confirmed on Zn added Ga_2O_3 if Ga_2O_3 which was prepared using dilute CaCl_2 solution was used. The photocatalytic activity in an inner-irradiation type photoreaction cell under irradiation using a high-pressure Hg lamp reached over 32 mmol h^{-1} for H_2 and 16 mmol h^{-1} for O_2 production, respectively, which was extremely higher than the photocatalytic activity we have reported previously. Then, the apparent quantum yield was evaluated under irradiation

of monochromatic light at 254 nm using a top irradiation type cell. The value over the $\text{Rh}_{0.5}\text{Cr}_{1.5}\text{O}_3/\text{Zn-Ga}_2\text{O}_3(\text{UP-Ca}(0.001))$ photocatalyst was 71%, while that over $\text{Rh}_{0.5}\text{Cr}_{1.5}\text{O}_3/\text{Zn-Ga}_2\text{O}_3(\text{DI})$ was 57%.

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